

AI-Driven Land Governance Intelligence: Integrating Heterogeneous Land Records and Public Grievances for Evidence-Based Policy Decision-Making

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Abstract

Land governance is a critical determinant of sustainable development, but it is often a disconnected set of registries, both within governments and across civil society. This includes land revenue departments, survey and settlement records, geospatial datasets, and citizen grievance forums, among others. By virtue of being soiled systems, these registries are unable to provide the holistic view needed to identify patterns such as clusters of particular land disputes, persistent processing delays in mutation or title regularization, or regions with particularly high concentrations of grievances. This paper attempts to address the research question of how to operationalize a significant amount of information digitized from land governance systems without adequate analytical tools to process them into useful insights for policymakers. It proposes an AI-assisted framework to analyze the patterns in land governance, primarily using structured data from various land records while also incorporating semi-structured data from surveys and unstructured complaints from citizen forums. The framework aims to collate these insights into a coherent body of information that can help identify areas of the land governance system requiring improvement.

First, the paper will harmonize the disparate datasets on land governance, including structured data items such as revenue, survey, and settlement records, and unstructured text items such as grievance complaints. It will attempt to utilize entity resolution and geospatial analysis to align the data items on a uniform framework. Second, it will analyze the data using natural language processing approaches to extract patterns and themes from the grievance complaints, identify relevant issues amenable to policy intervention, and highlight areas of concern for further investigation. Third, the paper will collate the findings from the previous two steps to highlight correlations, such as particular governance issues that fuel certain types of disputes or cause delays in processing.

The proposed framework for collecting and analyzing the datasets will be tested on a representative sample of data, either from existing open datasets or generated from first principles. Evaluation metrics will focus on accuracy in categorizing grievances, detecting anomalies, and extracting relevant insights for policymakers. By building an operational analytical framework, this paper aims to contribute to the solution of unlocking the insights contained within disparate digital assets related to land governance. The findings will enable targeted interventions in the areas of most urgent need and help prioritize future improvements. The study will also provide a replicable example of how artificial intelligence can be used in public administration to generate trustworthiness and transparency in governance.